SentraCore - Group 3
CSIT321 - Planning and Feasibility Report

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Background Information

Many students today face challenges when learning on their own. It can be hard to stay motivated, understand complex topics, or find the right support at the right time. Most online learning tools focus on individual study and don't offer much help with teamwork or real-life preparation.

We created SkillSync to solve this problem. It's a platform where students can learn together and support each other. Whether they're preparing for exams, building new skills, or practicing for interviews, students can work in goal-based study groups, connect with peers, and get live help when needed.

SkillSync also helps students get ready for their future careers with tools like AI-powered interview practice and hands-on challenges. By combining learning, teamwork, and career preparation, SkillSync makes studying more engaging, effective, and connected.

Project Description & Objectives

- **What is our Project?**

  Introducing **SkillSync** – Your perfect pathway to collaborative learning and career excellence!

  SkillSync is an innovative platform that reshapes the process of learning and preparing for next future steps. By integrating study pods that are goal-oriented, immediate, and interactive career readiness, SkillSync redefines learning into a shared experience. Whether you are building a new skill, preparing for examinations, or gearing up for job interview opportunities, SkillSync enables you to study smarter and accomplish greater success side by side.

  Discover **Collaborative goal-based study pods** to create mutual learning targets and tracking achievements as a team. Connect through **PeerLink** effortlessly to receive live assistance or share your expertise. Match with your ideal study companion with **Study Buddy Match**, and access instant assistance when it counts with **Quick Help Live Chat**. Elevate your interview preparations and tackle job-focused tasks with **AI Avatar** interviewers**.**

SkillSync unites the gap between learning and practical experience, empowering you to pursue your aspirations confidently. Through its seamless integration of study, teamwork, and career development, SkillSync guarantees that every step you make propels you toward your objectives. Join us in transforming the journey to learn, develop and thrive together.

- **Main Problems & How do we address them?**

  One of the biggest challenges in education today is that many students and self-learners struggle to stay motivated and grow effectively when learning on their own. It can be hard to stay consistent, have people to study with, or get feedback that is helpful. On top of that, there aren't many platforms that support both learning and career preparation in one place, leaving learners to rely on multiple apps and tools that often lack coordination and clear guidance.

  SkillSync addresses these issues by offering a structured, all-in-one platform. It offers four essential features that work together to aid learners:

  - **Collaborative Goal-Based Study Pods:** Study Pods are group sessions focused on a shared goal, such as completing a project, preparing for a test, or learning a new skill. Each pod has a specific deadline to reach the goal, such as "Prepare for the biology final exam in 3 weeks." Users have the option to start their own pod or join an existing one that aligns with their interests. The user who initiates the pod defines the topic, sets the deadline, and selects the preferred mode of collaboration, whether online, in-person, or hybrid.

    - Each pod includes a **Progress/Milestone Tracking** feature that allows users to set long-term objectives/goals and the system divides it into phases that the user can modify to suit their preferences. Users can check off their goals after they've been set, and see their progress displayed with a visual bar. A **shared whiteboard** lets members collaborate in real time, making it easy to brainstorm and share ideas. Additionally, users can stay organized by writing and saving notes in the **Note taking module**. The system prompts the user to either mark their notes as understood, review later, or add an update and additional reminders.

- **Peer Link – Peer-to-Peer Support Network:** PeerLink is a space where users can ask for help or offer assistance on specific topics, promoting peer-to-peer learning. Users can communicate through text chats by replying to the questions, or they can schedule meetings through the Study Pods to discuss topics in real time. To keep sessions comfortable and focused, camera use is disabled during meetings, but users can share their screens to better explain or demonstrate ideas.
    - PeerLink includes **Study Buddy Match**, which pairs users with study partners who share similar interests or goals. Users can connect with others through **Quick Help Live Chat** for fast, real-time support. Mentors mark themselves as “Available to Help” so users know they are available for live help. Additionally, a **shared whiteboard** allows participants to collaborate visually and solve problems together.
- **Peer Feedback System:** After receiving help through PeerLink or upon completing a study pod session, users can rate the people they worked with using a **5-star system** based on clarity, communication, and preparedness. The system then calculates an average rating, which is displayed on each user’s profile. These ratings contribute to a Reputation Score and assign tags based on the calculated average rating. For example, users with an average rating between 4.8 and 5.0 earn the "Top Mentor" tag, which is also shown on their profile.
- **Career Companion – Interview Preparation Toolkit:** Career Companion helps users prepare for real-world job interviews in a smart and engaging way. It features **AI Avatar Interviewers** that simulate realistic interview scenarios. Users begin by selecting the job role they want to practice for, such as Software Developer or Marketing Intern, and then review common questions for that role. Users can either record themselves answering the questions and upload the video for feedback, or choose an AI avatar to conduct the interview. Each avatar represents a different interviewer personality based on real-life experiences.
    - For example, users can practice with a strict Corporate HR Manager who asks formal questions like “What are your greatest strengths and weaknesses?”, a chill Startup Co-founder who asks creative questions like “What would you build if you had full freedom?”, or a

serious Technical Lead who asks detailed problem-solving questions like “How would you design a login system?”

- After each session, users receive **AI-powered feedback** on their answers, including suggestions to improve their tone, clarity, and confidence. This helps learners practice and build skills in a safe environment before facing real interviews.

Project Scope

- **Competitor solutions & their limitations**

  Interviewing.io:
  It is a well-established interview-readiness platform, it is known for facilitating anonymous practice interviews with engineers who are experienced from prominent technology companies. In this platform, it allows users to rehearse coding interviews in a realistic setting, providing instant feedback and detailed performance measurements. This offers a vital hands-on approach for enhancing confidence and perfecting problem-solving skills under stress (Interviewing.io, n.d.).

  Despite with its advantages, Interviewing.io faces few significant limitations as follows:

  - **Narrow Focus:** Mainly emphasizes on technical practice interviews for only software engineering positions, making minimal support for other fields of profession.
  - **Lack of Peer Engagement:** Fails to provide peer-led study pods, mutual goal planning, or group-based learning sessions.
  - **Absence of Skill Sharing:** Does not support real-time expertise sharing or team-based problem-solving beyond individual practice interviews.
  - **Weak Foundation for Long-term Learning:** Lacks mechanisms for structured learning journeys, ongoing project initiatives, or milestone-oriented progress monitoring.

- **Restricted Community Involvement:** Interaction mostly revolves around the practice interviews, offering minimal opportunity for community-based learning sessions or networking.
- **Feedback Lacks Depth:** Although the users receive evaluations for mock interviews, there is no build-in support for continuous peer reviews or reflective learning discussions in study pods.

This is where SkillSync marks as a key differentiator. Rather than focusing solely on interview-readiness such as Interviewing.io, it blends group-oriented learning via Study pods, live peer interaction through PeerLink, and community-based achievement tracking. With its goal-focused study teams and defined milestones, SkillSync connects the divide between self-study and group-based learning, promoting holistic development and an encouraging learning community.

Udemy:

It is a course hub that provides global online services to its users. Instructors create specialized topics that aid their consumers to learn as well as enhance their skills that pertain to their job or interest. The platform supports courses in several languages that cater to a great number of users worldwide. Other than individual users, the platform provides their services to businesses as well. Users can follow a personalized lesson to reach their learning goal better (Udemy, n.d.).

Despite with its advantages, Udemy faces few significant limitations as follows:

- **Course Quality:** The quality of the courses will vary depending on who is creating it and guiding the learners. In Udemy, anyone can create as well as publish the content of the course and this affects the overall experience for the users.
- **Course Recognition:** Even Though Udemy is a popular platform for learners to acquire or enhance a skill, the certificates received after learning that courses are not recognized officially or accredited by educational organizations.
- **Outdated Course Content:** As the amount of courses increases, the content quality checking services reduce drastically. This leads to users

learning outdated contents which may not cater to the growing needs of the current as well as upcoming generations.

- **Minimal Interactions:** The platform does provide interactions with instructors or other users taking the course at the same time. As most of the contents are created beforehand, there are no live interactions taking place although there is a discussion forum.

This is where SkillSync takes the effort to provide worthwhile sessions to their users. The contents are provided by the creators of the pods, which ensures that the course is up to date, and while the course is ongoing in the study pods, learners can collaborate and interact with each other. Other than study pods, the PeerLink features lets users get their doubts or enquiries cleared without waiting for long as the platform provides quick help chats with the mentors.

Study Together:

It is a global platform that provides users virtual video call rooms or focus rooms. Users can either join a global study room call with other users from around the globe or with their friends and complete their own goal during the time. Their aim is to create an environment which focuses on providing a motivating scene for users to complete their tasks (Study Together, n.d.).

Despite with its advantages, Study Together faces few significant limitations as follows:

- **Lack of guide:** Even though there are focus rooms and study rooms, users on call are not guided through trackers or milestones to reach their end goals.
- **Limited tools for skill building:** The platform does not focus on fostering the skills acquired through the tasks completed by the users during and after being in the rooms.
- **Reduced Efficiency:** Users often lose track of focus from their work when being in study rooms with other members and this can reduce the purpose

of the service. Users would be distracted and sessions have the possibility to turn into a socialisation point between peers or/and strangers.

- **No live feedback:** There is no system in the platform and provides users to get their work checked and critiqued on when they are in the rooms. This can lead the user to complete their task with errors and if there are users struggling, there is no way to talk and aid them at the moment.

This is where SkillSync takes the chance to implement a peer feedback system as well as progress trackers. The progress trackers are available for users to check in their study pods along with the course milestone, and this will aid learners to stay on the right path while still learning a new skill. In addition, users can collaborate with other learners in the PeerLink service where users can answer the doubt or hop on a quick session to understand and clear the doubt in a better format.

Brainly:

Brainly is a popular academic Q&A platform where students post questions and receive answers from peers, educators, or verified experts. It is well-suited for quick homework help and covers a broad range of school-level subjects.

Despite its advantages, Brainly comes with several limitations that restrict its effectiveness as a comprehensive learning platform as follows:

- **Absence of Group Learning or Collaboration:** Brainly only allows users to ask and answer queries. There are no group meetings or resources available for students to work together toward the same goal, like preparing for a test or doing a project.

- **Delayed Response Time:** On Brainly, users often have to wait for someone to notice and answer their questions. The platform doesn't offer real-time chat or immediate support for urgent help.

- **Limited Career Support:** Brainly mainly focuses on helping students with school subjects. It doesn't offer tools or features to help users with job skills or interview practice.

This is where SkillSync redefines the student learning experience. It enables group collaboration through goal-based study pods where learners work together toward shared deadlines. It also offers instant help via live chat and matches users with study buddies who share similar interests. SkillSync supports interview preparation through AI-powered mock interviews as well. Overall, SkillSync provides a complete platform for both learning and career growth.

- **Target Niche**

SkillSync is designed for students across all fields of study who want a more collaborative and engaging way to learn. It offers goal-based study pods, real-time peer support, and AI-powered interview practice, making it the ideal platform for students who want to study smarter, grow together, and feel confident in their academic and professional journey.

- **Prerequisites**

User Prerequisites

- **Access to a Compatible Device:** The user should have a laptop or phone that supports a current browser in order to access pods, share skills or practice interviews.
- **Stable Internet connection:** The user requires a stable internet connection for joining live sessions, receiving real-time support, collaborating with shared whiteboards, or utilising the built-in code editor.
- **Willingness to Learn Collaboratively:** Students must be open to working with their peers—whether in a study pod, giving or receiving feedback, or learning with study buddies—to fully leverage the social model of the platform.
- **Basic Familiarity with Learning Tools:** No particular technical skill is required, but users must be comfortable using basic tools like calendars, progress tracking, and communication tools.

Technical prerequisites

- **Goal and Pod Management system:** A backend mobile is needed to handle creation, tracking, and updates of user goals and study pods. This includes timeline, milestones, and collaboration settings.
- **Real Time Collaboration Infrastructure:** The system must support embedded tools such as live whiteboards and code editors, along with real time messaging or video options for smooth peer-to-peer interaction during study sessions.
- **Matching and Feedback Algorithms:** An algorithm is required for intelligent pairing of users based on availability, learning style, and goals, the feedback engine should compute reputation scores and assign community trust badges.
- **AI Powered interview Toolkit:** The interview module depends on AI models that simulate different interviewer persona, analyze user responses, and offer feedback on clarity, tone, and confidence in real-time.

- **Limitations**

While SkillSync offers a new approach for workforce preparation and collaborative learning, several constraints should be kept in mind at the planning and development stage.

- **Development Overhead and Technical Complexity:**
  SkillSync integrates several advanced features, including AI-driven mock interviewers, live chat, and tracking progress, all demanding sophisticated back-end infrastructure and seamless front-end user experiences. Development of such a multi-layered platform can lead to prolonged timelines, high development costs, and ongoing maintenance complications. Such tech requirements also call for an extremely highly skilled development team, influencing project viability in its early stages.
- **AI and Automation Shortcomings:**
  All three of the major features AI Avatar Interviewers, tone analysers, and

auto-feedback features, depend on artificial intelligence and natural language processing. While such technology holds much promise, it is still imperfect. It can misinterpret the user responses, the user attitude, or the user intention, providing incorrect feedback, influencing user trust. This is an area of risk, mostly for users who extensively use SkillSync for interview preparation or for assessing personal development.

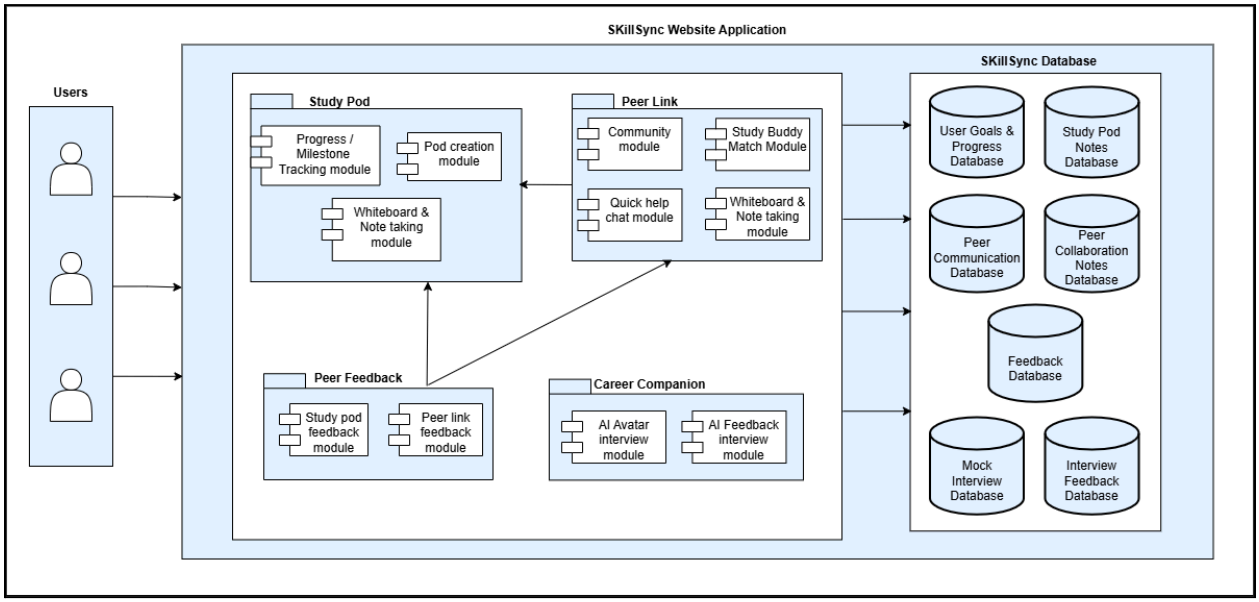
- **Over-reliance on Community Involvement:**
  One of SkillSync's most distinctive features is its peer-leadership focus through PeerLink, Study Pods, and Study Buddy Match. The success of such features, however, depends largely on ongoing user engagement and the availability of enough active users. In the absence of early aggressive adoption, together with user motivation, pods may remain idle, and the support community may disintegrate, reducing the site's value proposition.

- **Scalability, Accessibility, and Resource Constraints:**
  In the initial stage, the platform may not be scalable, particularly for high user load or real-time co-editing scenarios like video sessions and pair programming. Moreover, its access for users with disabilities or low-bandwidth locations may be difficult if not addressed in the early stages of development. Therefore, its adoption may be limited in the initial stages to tech-savvy, English-speaking individuals, affecting reach and inclusivity.

- **Moderation, Quality Control, and Trust Issues:**
  Since the forum encourages open peer-to-peer collaboration and knowledge-sharing capabilities, quality control among the information contributed, comments, and behaviour will not be easy. Without stringent moderation features and clear community guidelines, there is potential for misinformation, inconsistent help, or even negative user experiences. Safety, productivity, and dependability will need to be provided then, though at the expense.

How do we do it?

- **High-level System Architecture**



- **High-level System Operation**

  - **Users**
    - The leftmost section shows **Users** who interact with the system.
    - They access different features of the application via various modules.

  - **SkillSync Website Application**
    This is the main body of the application, containing four main functional areas:

    - **Study Pod**
      Focuses on learning and progress tracking:
      - **Pod Creation module:** Handles the setup of new study pods, including goal, deadline, and collaboration mode.

- **User Progress/Milestone Tracking module:** Allows users to set learning goals and track learning milestones.
- **Whiteboard and Note-taking module:** It provides a space for sharing ideas, taking structured notes, and supporting visual learning for an easier learning experience.

- **Peer Link**

  Facilitates peer-to-peer learning and community interaction:
  - **Community module**: Connects users with a broader learning community.
  - **Study Buddy Match Module**: Matches users with study partners who share similar interests.
  - **Quick Help Chat Module**: Enables real-time peer support for immediate assistance.
  - **Whiteboard and Note-taking Module**: It provides a space for sharing ideas, taking structured notes, and supporting visual learning for an easier learning experience.

- **Peer Feedback**

  Provides structured peer review and feedback across modules:
  - **Study Pod Feedback Module**: Users rate peers after Study Pod sessions.
  - **Peer Link Feedback Module**: Users rate peers after receiving help through Peer Link.

- **Career Companion**

  Supports career preparation through AI-based tools:
  - **AI Avatar Interview Module**: Simulates mock interviews with AI avatars.
  - **AI Feedback Interview Module**: Analyzes and provides feedback on recorded interviews.

- **SkillSync Database**
  Stores and manages all data used across modules:
    - **User Goals and Progress Database**: Stores pod goals and milestone progress information.
    - **Study Pod Notes Database**: Saves individual and shared notes from Study Pod sessions.
    - **Peer Communication Database**: Logs messages and community interactions.
    - **Peer Collaboration Notes Database**: Stores shared whiteboard content from collaborative sessions.
    - **Feedback Database**: Centralizes all user ratings from Study Pods and Peer Link sessions.
    - **Mock Interview Database**: Records details of mock interview sessions with AI avatars.
    - **Interview Feedback Database**: Stores AI-generated feedback from user interviews.

- **User Scenario (System flow)**
    - **User Login & Navigation:**
        - A user logs into the SkillSync web application.
        - The user is taken to the dashboard and chooses to join or initiate a Study Pod.
    - **Study Pod Setup & Engagement:**
        - The user sets learning objectives with the Module-based User Goal Creation tool.
        - Such objectives are stored in the User Goals & Progress Database.
        - The user learns, he/she can:

        - Track milestones through the Progress/Milestone Tracking Module.
        - Can form or join study groups using the Pod Creation Module
        - Collaborative notes can be taken using the Whiteboard & Note-taking Module
        - Feedback is exchanged with members of a pod via the Peer Feedback Module
        - Post Link Feedback Module provides feedback after a meeting
        - Ongoing peer review takes place via the Peer Link Feedback Module
    - Information is being stored in the Study Pod Notes and Peer Collaboration Notes Database for shared edits, and the Peer Feedback Database for pod feedback logs. Data collected from tracking goals and users' progress as part of the Study Pod system will be stored in the User Goals & Progress database.

- **Peer Collaboration via Peer Link:**
    - The user logs in to the Peer Link module to collaborate.
    - They can:
        - Chat through the Quick Help Chat Module.
        - Match partners with the Study Buddy Match Module.
        - Collaborate on a shared Whiteboard & Note-taking Module.
    - Notes and communication are saved in the Peer Communication and Peer Collaboration Notes Databases.

- **Getting Feedback:**
    - The user gives or receives feedback in three places:
        - Study Pod Feedback Module
        - Peer Link Feedback Module
    - Feedback is all held in the central Feedback Database.

- **Career Preparation via Career Companion:**
    - The user logs in to the AI Avatar Interview Module or AI Feedback Interview Module for mock interviews.
    - Interview attempts and AI feedback are saved in the Mock Interview and Interview Feedback Databases.

- **Backend Operations:**
    - During the flow, the web application interacts with the SkillSync Database to read/write:
        - Goals and Progress
        - Notes and chats
        - Feedback and mock interview data

Software and Hardware Requirements

- **Software Requirements**
    - **Tools & Platform**
        - **Frontend**:
            - **React.js**: This tool will be used to create interactive components for all the user interface features in the project.
            - **Tailwind CSS**: This tool will be used to implement the design of the interface in the project.

    - **FullCalendar.js**: This tool will be used to display timelines, sessions that can or are scheduled and goal timelines in the study pods.

- **Backend**:
    - **Node.js**: This tool will be used to handle as well as run the backend real-time processes efficiently.
    - **Express.js**: Server routing for managing the APIs such as the career companion interview data, study pods greetings, etc.

- **Automation**:
    - **n8n**: This tool will be used to send reminders to users, health related reminders about the pods, feedback alerts, etc.

- **Authentication**:
    - **Firebase Authentication**: This tool will handle the secure sign-ins which will be role based (mentor, learner, peer,etc) and will handle user sign ups. It will also manage the sessions taking place for the pods, career companions and PeerLink features.

- **Database and Storage**:
    - **MongoDB**: This tool will store all the data regarding the users such as progresses, goals, timelines, interview data, etc.
    - **Firebase Storage**: This tool stores data such as the mock interviews and avatar videos that will be used for the interview.

- **AI Avatar**:
    - **HeyGen**: This platform will be used to create different personality talking AI avatars that will be used in mock interviews for users to practise on.

- **OpenAI/Chatgpt**: This tool will be used to create the dialogues that would potentially be added into the avatars.

- **AI Integration:**
  - **AssemblyAI:** This tool will be used to transcribe the mock interview that took place into text and will be used to analyze and give feedback.
  - **Hugging Face sentiment analysis**: Once the meeting is transcribed, the tone/wordings will be analysed using this tool.
  - **Bubble.io**: This platform will be used to build the website application with AI plugins.

- **Authentication:**
  - **Firebase Authentication**: This tool will control as well as manage the logins in a secure manner. Other than login, it will handle session controls such as the access for the mentors and learners.

- **Communication:**
  - **WebRTC**: This tool will allow the users to interact with the AI Avatar or general aspects that are in interviews that take place whether it be audio or video.
  - **Recorder.js**: This tool records the interview sessions, so that the user can replay their video for analysis and interview purposes.
  - **Google Meet**: This platform will be implemented to allow the video/audio call features that are common in all of the main features in this project.

- **Design:**
  - **Figma**: This platform will be used to create the design for the high fidelity prototype of the project before the actual

implementation. It will be used to map the UI/UX design of the system.

- **Draw.io**: This platform will be used to create the high level architecture design, to better visual how the system will be connected as well as working.

- **Testing:**
  - **Postman**: This tool will be used to test the versions and the units and integration for the features of the project such as pod creation, study buddy match and so on.
  - **GitHub**: This tool will be used to manage versions along with other security and system testing as well.
  - **Firebase Security Rules testing**: This tool will be used to test the security of the firebase authentication as well as storage.
  - **React testing library**: This tool will be used to test frontend components of the project to test the UI as well as user behaviour.

- **Hardware Requirements**
  - **System Setup Hardware**
    - **Setup from Server-Side**
      - **Cloud Infrastructure (Google Cloud/AWS):** This is utilized for deploying backend services, to ensure robust availability, seamless performance, and automatic scalability under peak usage.
      - **Real-Time Databases (MongoDB/Firebase):** This handles dynamic content such as the study groups, chat, live sessions, and immediate user updates with safety and efficiency.
      - **Load Balancers:** Allocating interaction of users across multiple server clusters to prevent the risk of overloading and ensure stable uptime and performance.

- Minimum Requirements from Client-Side
  - **Memory (RAM):** Not less than 4GB. This is to facilitate multiple tasks simultaneously during virtual meetings, live sessions, and collaborative workflows.
  - **Processor:** Dual-core of 1.5 GHz or higher. This is sufficient for handling modern web interfaces and multimedia content run smoothly.
  - **Storage space:** Requires 128GB (with 2GB available). This is necessary to store temporary files, local cache, and session logs to ensure a smooth user journey.
  - **Speed of Network:** Not less than 10 Mbps. This is crucial for an uninterrupted video chats, live document editing, and interactive learning experience.
  - **Browser Compatibility:** This is tailored for major web-browsers such as Chrome, Safari, and Firefox, and also for mobile systems such as iOS and Android for convenience and accessibility.

- Technologies for Performance & Communication
  - **WebSocket Integration:** This supports real-time user communication with minimal delay, which is essential for prompt feedback and collaboration.
  - **Video Conferencing APIs (WebRTC):** This promotes a smooth video and audio calling feature for PeerLink and Study Pods sessions within the application, improving an interactive learning environment.

- **Specialized Hardware**

  Our platform works through a website, so most people can use it with a regular laptop or phone. But for some of the advanced features, we need a few special hardware tools, mainly on the server side.

  - **For the Server (backend system):**
    - **Graphics Card (GPU):** This is needed for the AI avatars in the Career Companion feature. It helps the system run the avatars and give feedback quickly and smoothly.
    - **Cloud Hosting with Good Performance:** We need strong cloud servers like AWS or Google Cloud to support many users at the same time. This helps the website stay fast during live chats, whiteboard sessions, and AI interviews.

  - **For Users (optional but helpful):**
    - **Headset with Microphone:** Useful for better sound quality during live chat or study sessions.
    - **Tablet with Stylus:** Makes it easier to draw or write on the shared whiteboard, especially for subjects like math or design.

Challenges

One of the main challenges in building SkillSync is creating a technically robust platform that remains intuitive and engaging for users. Features like Avatar Interviewers and PeerLink rely on seamless integration of AI and real-time communication, which can be difficult to maintain at scale. Another challenge lies in keeping users consistently involved, ensuring that study groups are active and peer interactions are meaningful and requires thoughtful design and moderation. Addressing these hurdles involves not just advanced technical skills, but also a deep understanding of user behavior. With the right balance, SkillSync has the potential to become a reliable, collaborative space for continuous learning.

Future Improvements

Due to time and scope constraints, several advanced features could not be included in the initial release. The following ideas are proposed as possible future additions that could be developed in follow-up stages or as upgrades depending on demand and platform evolution:

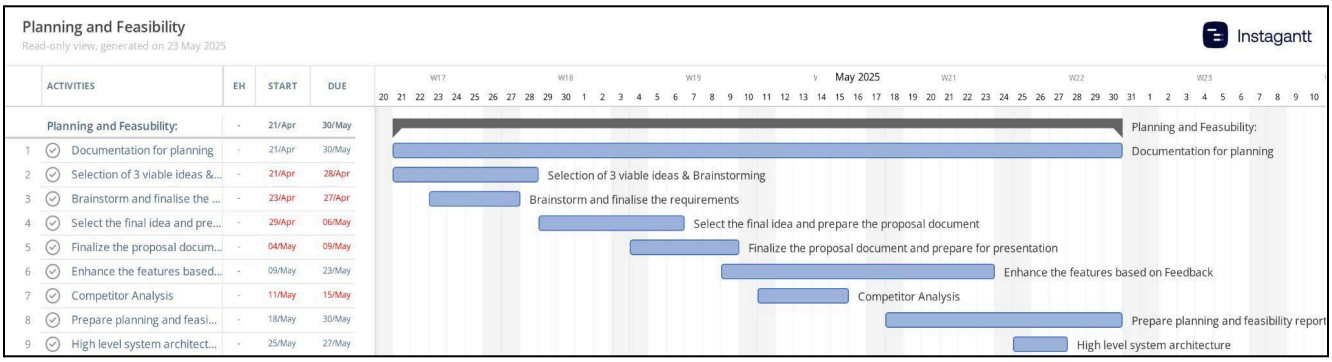
- **Pod Health Monitor:** To ensure users stay active and encourage active interaction, the feature would provide visual indicators of Study Pod activity. Numbers such as task completion, check-in rates, and discussion activity would allow idle pods to be identified and initiate re-engagement activities.
- **Live Code Editor for Study Pods & PeerLink:** Future versions could include programming-specific Study Pods and PeerLink sessions with a collaborative code editor embedded in them. This would allow for real-time pair programming, debugging, and code reviews in the platform, reducing third-party tool reliance.
- **Skill Synergy Challenges-Role-Based Mini-Games:** These mini-games would mimic actual work scenarios, such as on-the-spot debugging, marketing pitching, or design mock-up commenting. Although interesting and useful, more complex and will be introduced following the main learning elements' maturation.
- **"Time Warp Debrief" AI-Powered rewind:** Following mock interviews, users may view highlights and receive feedback powered by AI. This includes suggested alternative responses, tone/pace guidance, and body language signals would all be provided to help users reflect and improve.

Timeline - Gantt Chart

The timeline for this gantt chart starts from April 21st up until Dec 7th. Due to this long time period and the timeline consisting of a good number of phases, the gantt charts have been broken down into separate charts with phases, but to better know the timeline, the dates are presented beside each phase.

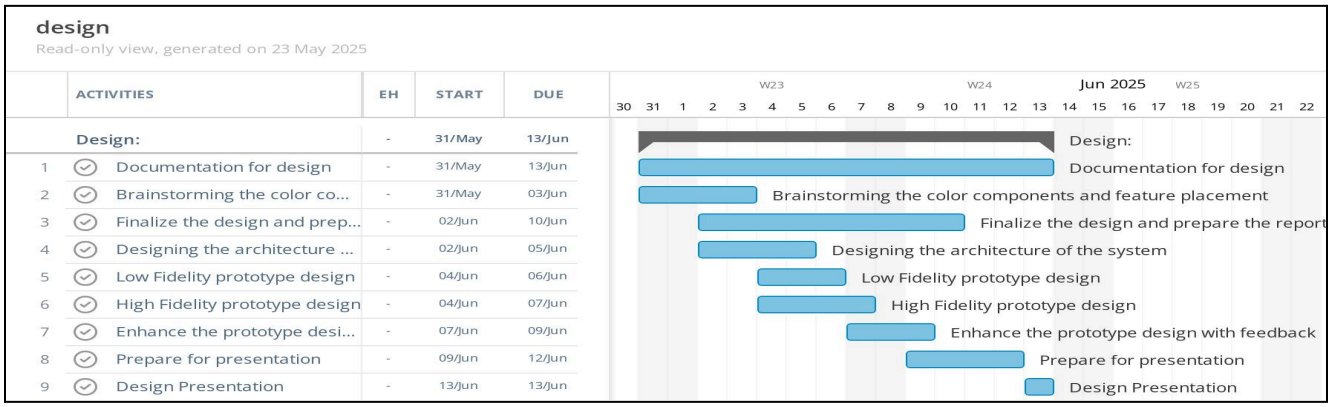
The timeline spans to a total of 588 days. The below table shows the breakdown of the gantt chart:

Task Name	Start Date	End Date	Duration (Days)
Planning and Requirements			
Documentation for planning	21/04/2025	30/05/25	40
Selection of 3 viable ideas & Brainstorming	21/04/2025	28/04/25	8
Brainstorm and finalise the requirements	23/04/2025	27/04/25	5
Select the final idea and prepare the proposal document	29/04/25	6/5/2025	8
Finalize the proposal document and prepare for presentation	4/5/2025	9/5/2025	6
Enhance the features based on Feedback	9/5/2025	23/05/25	15
Competitor Analysis	11/5/2025	15/05/25	5
Prepare planning and feasibility report	18/05/25	30/05/25	13
High level system architecture	25/05/25	27/05/25	3



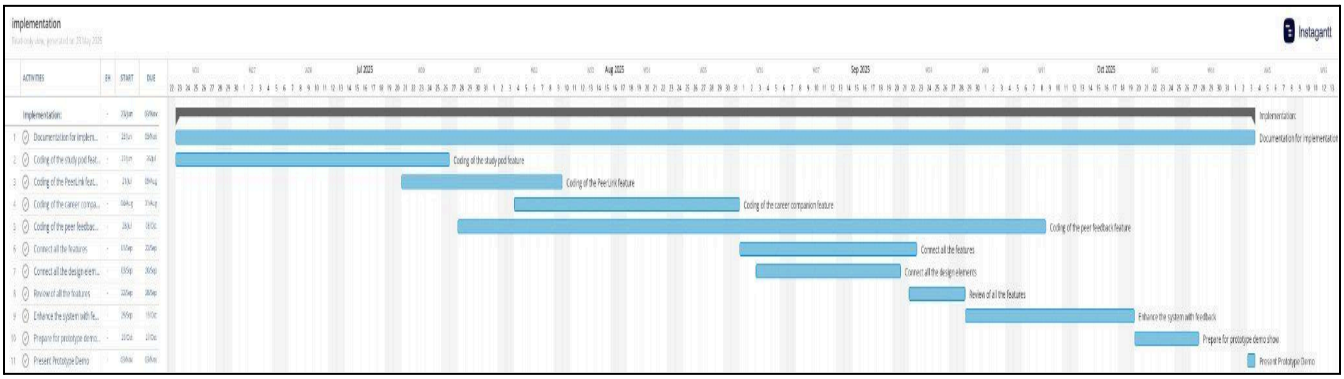
Task Name	Start Date	End Date	Duration (Days)
Design			
Documentation for design	31/05/25	13/06/25	14
Brainstorming the color components and feature placement	31/05/25	3/6/2025	4

Finalize the design and prepare the report	2/6/2025	10/6/2025	9
Designing the architecture of the system	2/6/2025	5/6/2025	4
Low Fidelity prototype design	4/6/2025	6/6/2025	3
High Fidelity prototype design	4/6/2025	7/6/2025	4
Enhance the prototype design with feedback	7/6/2025	9/6/2025	3
Prepare for presentation	9/6/2025	12/6/2025	4
Final Presentation	13/06/25	13/06/25	1

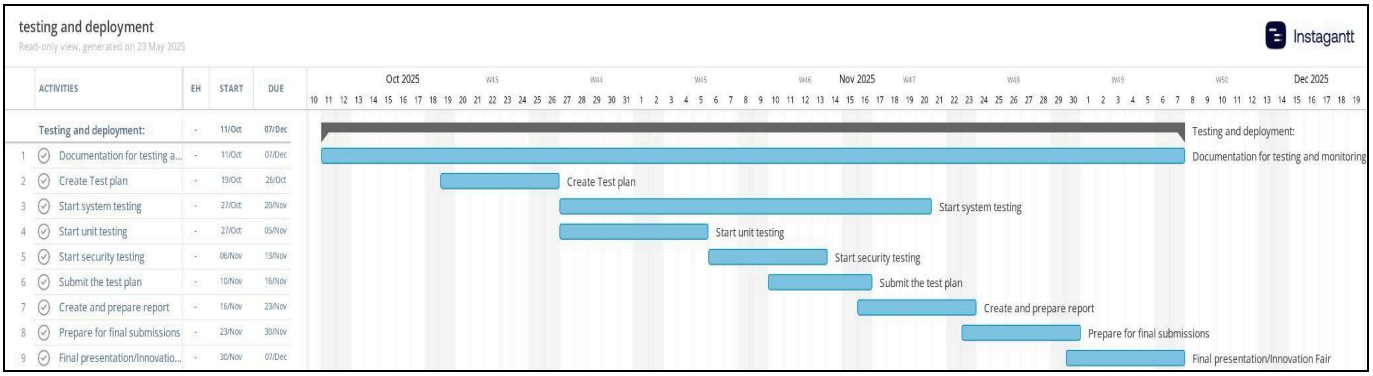


Task Name	Start Date	End Date	Duration (Days)
Implementation			
Documentation for implementation	23/06/25	3/11/2025	134
Coding of the study pod feature	23/06/25	26/07/25	34
Coding of the PeerLink feature	21/07/25	9/8/2025	20
Coding of the peer feedback feature	28/07/25	10/8/2025	14
Coding of the career companion feature	4/8/2025	31/08/25	28
Connect all the features	1/9/2025	22/09/25	22
Connect all the design elements	3/9/2025	20/09/25	18
Review of all the features	22/09/25	28/09/25	7

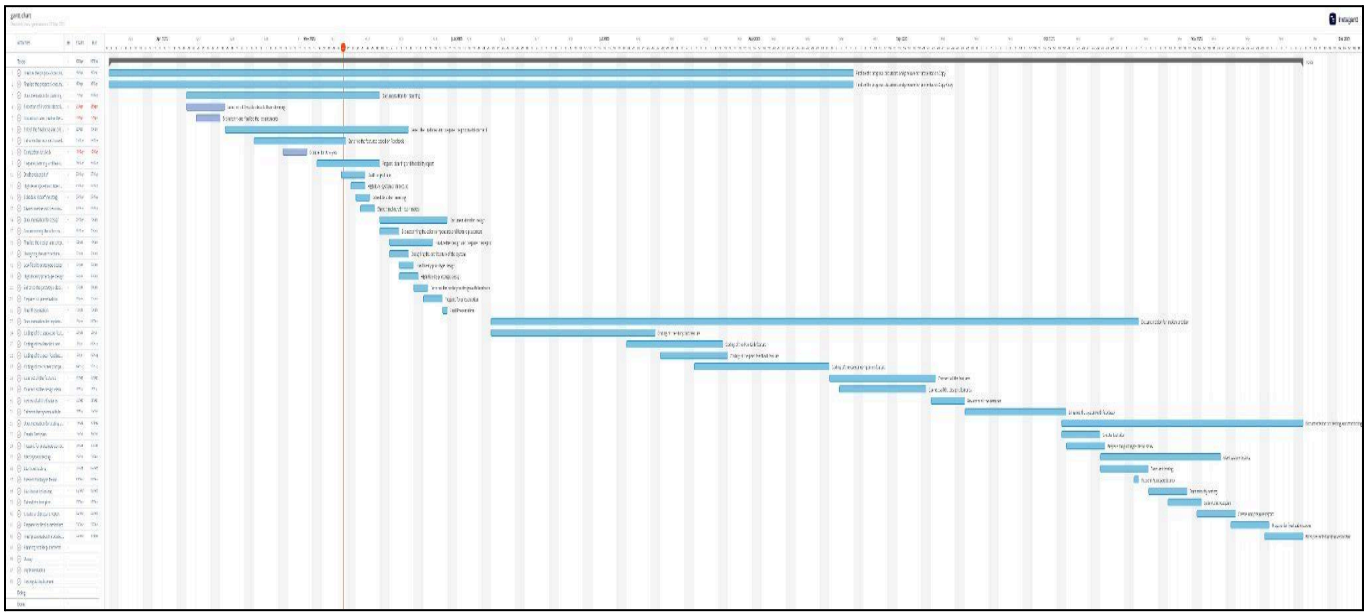
Enhance the system with feedback	29/09/25	19/10/25	21
Prepare for prototype demo show	20/10/25	27/10/25	8
Present Prototype Demo	3/11/2025	3/11/2025	1



Task Name	Start Date	End Date	Duration (Days)
Testing & Deployment			
Documentation for testing and monitoring	19/10/25	7/12/2025	50
Create Test plan	19/10/25	26/10/25	8
Start system testing	27/10/25	20/11/25	25
Start unit testing	27/10/25	5/11/2025	10
Start security testing	6/11/2025	13/11/25	8
Submit the test plan	10/11/2025	16/11/25	7
Create and prepare report	16/11/25	23/11/25	8
Prepare for final submissions	23/11/25	30/11/25	8
Final presentation/Innovation Fair	30/11/25	7/12/2025	8



Below shows the expanded gantt chart view with all the components combined:



Project Management tools to be used

The following are the key tools and platforms that are used to ensure our project progresses according to the work distribution and planned timeline:

- **Google Document:** Utilized as the primary tool for managing the documentation part of our project. With its real-time collaboration functionality enabled all our teammates to work together at the same time, track the edits made, and ensure that each team member was contributing to their assigned parts. This enhanced responsibility and transparency across the team.

- **Google Meet:** Used for hosting online meetings when in-person meetings were not convenient. This helped to maintain clear communication among the team members and facilitated smooth decision-making on key project matters.
- **GitHub:** Essential for management of source code and version control. It enables collaborative programming through code integration, branch handling, and review workflows, which promotes a codebase that remains clear, efficient and streamlined.
- **Excalidraw:** This helps with collaborative whiteboarding, brainstorming, and visually mapping out user flows, structure of processes, and design of interfaces, which transforms vague ideas more concrete.
- **SmartDraw:** This website was utilized to create a structured and clear timeline for our project, which helps us as a team to visualize the estimated timeframe of each phase and ensure improved planning and allocation of time for the overall project.
- **Instagantt & Asana:** Both were used to make the Gantt charts that helped us to assess the tasks, plan the deadlines for each assigned tasks, and determine what was necessary to be completed to remain organized and fulfill our project goals.
- **Draw.io:** This was used to outline the High-Level Architecture of our application, which enabled us to visualise the overall structure of the system more clearly.

Team Work Distribution

Report Parts	Student Name	Student ID
Background Information	Yaxin Wang	7741091
Project Description & Objectives		
What is our Project?	Sharen Thomas	7700787
Main Problems & How do we address them?	Sara Ranjbar	7817848
Project Scope		
Competitor Solutions & their Limitations	Sharen Thomas Sikha Nair Sara Ranjbar	7700787 7810398 7817848
Target Niche	Yaxin Wang	7741091
Pre-requisites	Yaxin Wang Cadence D'Souza	7741091 7769714
Limitations	Cadence D'Souza	7769714
How do we do it?		
High level system architecture	Sharen Thomas Sikha Nair Yaxin Wang	7700787 7810398 7741091
High level system operation	Sara Ranjbar Cadence D'Souza	7817848 7769714
Software & Hardware Requirements		
Software Req - Tools & platforms	Sikha Nair	7810398
Hardware Req- System setup hardware	Sharen Thomas	7700787
Hardware Req - Specialized hardware	Sara Ranjbar	7817848
Challenges	Yaxin Wang	7741091
Future Improvements	Cadence D'Souza	7769714

Timeline - Gantt Chart	Sikha Nair	7810398
Project Management Tools to be used	Sharen Thomas	7700787
References	Sara Ranjbar Sikha Nair Sharen Thomas	7817848 7810398 7700787

References

Everyone

Interviewing.io. (n.d.). *Anonymous mock interviews with engineers from FAANG and more*. [online] Available at: https://interviewing.io.

Study Together (n.d.). *About | 24/7 Study Room & Focus Room* . [online] www.studytogether.com. Available at: https://www.studytogether.com/.

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